

Master's Programme in Life Science Technologies

Impact of sparse annotation and annotation noise on CNN-based medical image segmentation

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Abstract

Medical image segmentation plays an essential role in many clinical applications, including diagnostics and tumor growth monitoring. Convolutional neural network-based segmentation models have made segmentation more efficient, but training them requires a large amount of annotated data. Manual annotation is one of the major challenges in developing medical image segmentation models, as producing high-quality annotations requires considerable time and expertise from clinical experts. If training the models could be done with fewer annotations or annotations could be made faster by reduced annotation effort, it would make clinical applications more approachable.

In this research, the impact of sparse annotation and annotation noise on CNN-based medical image segmentation was studied with nnU-Net, the current state-of-the-art model for medical image segmentation, and the public KiTS23 dataset with CT volumes. To simplify the segmentation task, only single-class, kidney and masses, were segmented. To create the sparse-annotated and noise-added datasets, the codebase was developed as part of the research. To facilitate the analysis, a script was developed to automate the calculation of the evaluation metrics. The impact of sparse annotation was studied using three datasets in which 25% of voxels were annotated with different strategies, and two datasets in which 50% of voxels were annotated. All sparsifying was done in the axial direction. The impact of annotation noise was studied with one dataset, simulating annotations made by a rushed annotator, using elastic warping.

Because only one large organ was segmented, the models demonstrated strong segmentation performance, achieving high Dice scores across all datasets. With sparse annotations, false positives were observed when only 25% of the voxels were annotated. Moreover, the HD95 results suggest that annotation noise degrades boundary fidelity. The median HD95 in the noise-perturbed dataset is higher than in most sparsely annotated datasets. By contrast, the mean HD95 shows a smaller difference, likely because false positives in the sparsely annotated datasets inflate the mean.

Keywords sparse annotation, medical image segmentation, annotation noise, nnU-Net

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Lääketieteellisten kuvien segmentointi on tärkeässä osassa monissa kliinisissä sovelluksissa, kuten diagnostiikassa ja kasvaimen kasvun seurannassa. Konvoluutioneuroverkkoihin pohjautuvat segmentointimallit ovat tehneet segmentoinnista tehokkaampaa, mutta mallien kouluttaminen vaatii suuria määriä annotoitua dataa. Manuaalinen annotointi onkin yksi suurimmista haasteista lääketieteellisten kuvasegmentointimallien kehityksessä, sillä korkealaatuisten annotaatioiden tekeminen vaatii asiantuntijoilta huomattavasti aikaa ja osaamista. Jos mallien kouluttaminen voitaisiin toteuttaa vähemmällä annotaatioilla tai annotaatiot voitaisiin tehdä nopeammin vähentämällä työn määrää, se tekisi kliiniset sovellukset helpommin lähestyttäviksi.

Tässä työssä harva-alaisten annotaatioiden ja annotaatiokohinan vaikutusta CNN-pohjaiseen lääketieteelliseen kuva-segmentointiin tutkittiin käyttäen nnU-Net-ympäristöä, nykyisen johtavan tason mallia lääketieteellisten kuvien segmentoinnissa, ja julkista KiTS23-datasettiä, joka koostuu TT-tilavuuskuvista. Segmentoinnin yksinkertaistamiseksi segmentointiin vain yksi luokka, munuainen, ja muutoksenalueet. Harva-alaisesti annotoitujen ja kohina lisättyjen datasettien luomiseksi koodikanta kehitettiin osana tutkimusta. Tulosten analyysin sujuvoittamiseksi kehitettiin ohjelmakoodi, joka automatisoi arviointiparametrien laskemisen. Harva-alaisten annotaatioiden vaikutusta tutkittiin kolmella datasetillä, joissa 25% vokseleista oli annotoitu erilaisilla strategioilla ja kahdella datasetillä, joissa 50% vokseleista oli annotoitu. Kaikki harvennukset tehtiin aksiaaliseen suuntaan. Annotaatiokohinan vaikutusta tutkittiin yhdellä datasetillä, joka simuloi kiireessä tehtyjä annotaatioita käyttäen elastista vääristämistä.

Koska vain yksi suuri elin segmentoitiin, mallit osoittivat vahvaa segmentointisuorituskykyä ja saavuttivat korkeat Dice-kertoimet kaikissa aineistoissa. Harva-alaisilla annotaatiolla valespositiiveja havaittiin, kun vain 25% vokseleista oli annotoitu. Lisäksi HD95-tulokset viittaavat siihen, että annotaatiokohina vähentää rajojen tarkkuutta, sillä mediaani-HD95-arvo oli korkeampi kuin suurimmassa osassa harva-alaisesti annotoituja aineistoja. HD95 keskiarvo puolestaan ei ole niin korkea, todennäköisimmin johtuen valespositiiveista harva-alaisesti annotoiduissa aineistoissa.

Avainsanat harva-alainen annotaatio, lääketieteellinen kuvasegmentointi, annotaatiokohina, nnU-Net

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Symbols and abbreviations

Symbols

α	magnitude of warp
σ	granularity
n_{pred}	number of predicted pixels
n_{ref}	number of pixels in ground truth

Abbreviations

CNN	convolutional neural network
MRI	magnetic resonance imaging
nnU-Net	no-new-U-Net
CT	computed tomography
Dice	dice similarity coefficient
IoU	Jaccard index
TP	true positives
TN	true negatives
FP	false positives
FN	false negatives
HD95	95th percentile Hausdorff distance

1 Introduction

Medical image segmentation transforms the raw image data into structured information that is easier to analyze [1]. It plays an essential role in many clinical applications, like diagnostics and tumor growth monitoring [2]. Convolutional Neural Network -based segmentation models have made segmentation more efficient, but the generation of annotated datasets is one of the primary challenges in developing clinical applications, as the annotation process is labor-intensive and requires expert knowledge. [3].

Annotation is an important part of CNN-based medical image segmentation. It requires significant effort from experts, which makes the development of clinical applications usually inefficient and time-consuming [3]. If segmentation models could be trained on sparsely annotated or noisier data, it would make training more efficient and thus make clinical applications more approachable. The impact of sparse annotation has been researched several times [3],[4],[5], and strategies for sparse annotation have been researched for cardiac MRI [6], where the goal was to find out which slices of cardiac MRI are the most important to annotate. The impact of annotation noise has also been researched [7], where noisy labels worsened segmentation if added in over 20 % of cases. To the best of our knowledge, the impact of annotation noise has not been compared with sparse annotation, and thus, the optimum way to speed up annotation has not been concluded before.

The objective of this assignment is to design and implement a Python-based framework that enables systematic configuration and generation of datasets with controlled levels of annotation sparsity and label noise, in order to rigorously evaluate their impact on the performance and robustness of medical image segmentation models. A public dataset is used, and the models are trained using the nnU-Net environment.

In this work, section 2 discusses CNN-based medical image segmentation and evaluation metrics, section 3 discusses research material and methods, in section 4, results are shown, and last in section 5 are the conclusions.

2 CNN-based medical image segmentation

2.1 Annotations and strategies

For training the CNN-based segmentation model, one needs an annotated dataset. In the case of medical image segmentation, this often involves a lot of carefully crafted segmentation by experts, which takes hours and makes the development of clinical applications inefficient, time-consuming, and expensive [3]. If the work needed for annotation making could be reduced without a critical drop in models' performance, it would make the clinical applications more approachable. One approach for that is a sparse annotated dataset, where some voxels are left unannotated, or the annotation quality is reduced to speed up annotation.

The impact of sparse annotation has been researched before [6] with short-axis cardiac MRI segmentation. The research focused on which slices of MRI volumes would be the most important to annotate. For the general sparse annotation strategy, the study found that annotating more slices than volumes is a better strategy. [6]

The impact of annotation noise has been studied before [7]. The research used mandible segmentations from CT images and modified the ratios of noisy labels in the training set. The study found that noisy labels worsened segmentation, but when using 20% or less noisy cases for training, there was no significant effect. The study also suggests that the involved network is robust to noisy annotation to some extent, but highlights the importance of labeling quality. [7]

Realistic annotation noise is added before with elastic warping [8]. The warping is determined by a set of parameters: the magnitude of the warp (α) and the granularity (σ). The localized boundary ambiguity was simulated using a set of parameters, $\alpha \in [1, 15]$, $\sigma \in [5, 10]$. To simulate severe annotator inconsistency and coarse boundary delineation, an increased deformation magnitude was used while constricting spatial smoothness, setting $\alpha \in [30, 50]$, $\sigma \in [3, 5]$. For simulating shaking hands, one would use the first set of parameters, and for simulating a rushed annotator, the second set of parameters.[8]

2.2 nnU-Net

The nnU-Net is a Convolutional Neural Network (CNN)-based framework for medical image segmentation. It has achieved state-of-the-art performance across a wide range of datasets. The nnU-Net automatically adapts preprocessing, network architecture, training strategies, and post-processing to the properties of the given dataset. nnU-Net makes and trains by default three different configurations: 2D, 3D with full image resolution, and 3D cascade. Then it performs cross-validation and determines the best-performing configuration. The output is fully trained models ready for use. For further use, it is recommended to use an ensemble of all folds for the best results. [9]

The nnU-Net has an ignore label feature that can be used to mark regions that should be ignored during training. This is accomplished by using partial losses such that only non-ignored pixels are taken into account in training. During prediction and validation, nnU-Net predicts dense segmentation even when using the ignore label,

but during validation, only annotated pixels are used to compute metrics.[9] [10]

The nnU-Net gives for each fold the class-specific means and foreground mean that tells the model's average performance across all labels, excluding the background. All those means have the following parameters based on the confusion matrix: the dice similarity coefficient (DSC), the Jaccard index (IoU), True positives (TP) and negatives (TN), false positives (FP) and negatives (FN), the number of voxels the model predicted for that class (n_{pred}), and the number of pixels actually present in the Ground Truth (n_{ref}). [9]

2.3 Evaluation metrics

The Dice Similarity Coefficient (Dice)

The dice similarity coefficient quantifies the overlap between the predicted segmentation and the ground truth. The equation is given in 1, where P and Q denote the ground-truth and predicted masks, respectively. [11]

$$\text{Dice} = \frac{2|P \cap Q|}{|P| + |Q|} \quad (1)$$

The value of DSC is in the range of 0 to 1, where 0 means no overlap at all and 1 means perfect agreement between ground-truth and predicted mask [12].

The Jaccard Index (IoU)

The Jaccard index, also known as intersection over union, measures the similarity between predicted and ground-truth masks. The equation is given in 2. [11]

$$\text{IoU} = \frac{|P \cap Q|}{|P \cup Q|} \quad (2)$$

The value of IoU ranges from 0 to 1. The higher the score, the better. The IoU is a stricter measurement than DSC, and it penalizes false positives and negatives equally, whereas DSC gives more weight to shared positives. [13] Due to the similarity with Dice, only Dice is used in this work.

The 95th percentile Hausdorff Distance (HD95)

The 95th percentile Hausdorff Distance is defined as the 95th percentile of the shortest distances between boundary points of the predicted and ground-truth masks. It indicates the consistency of boundary accuracy; the lower the value, the better the results. [11]

Precision, recall & volumetric ratio

The number of true and false positives and negatives tells whether the model is over- or undersegmenting. For easier analysis, one can calculate precision, recall, and volumetric ratio. The definition of precision is shown in equation 3 and recall is shown

in equation 4 [14]. Volumetric ratio is the ratio between the predicted volume and the ground truth volume.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4)$$

3 Research material and methods

3.1 Dataset and compared scenarios

Dataset

The dataset from the 2023 kidney tumor segmentation challenge was chosen for this project. The dataset consists of 489 training cases comprising 3D CT images and segmentations, with background, kidney, tumor, and cyst labels, subsequently referred to as the KiTS23 dataset. The dataset’s parameters are shown in Table 1. [15]

Table 1: Characteristics of the KiTS23 dataset. Continuous variables are reported as median (range), and categorical variables as n (%).

Parameter	
Age (in years)	59 (18-90)
Gender (%)	63 M 37 F
Slice thickness (mm)	3.0 (0.5-5.0)

Due to the dataset’s large size, 100 cases were selected for actual research as the training set and 20 cases as the test set. The sets were chosen randomly and the same sets were used for all scenarios. The original dataset contains multiple regional labels, including separate annotations for tumor and cyst. In this work, only the original kidney and masses -label was used, thereby formulating the task as single-class segmentation rather than multi-class segmentation for simplification.

Baseline

The first dataset is a baseline in which the training cases are taken as they appear in the KiTS23 dataset. This dataset was used as the baseline, providing a consistent reference for meaningful comparisons across experiments.

The data processing pipeline

The codebase for creating sparse-annotated datasets and datasets with added annotation noise was developed for this research, as was code for calculating evaluation metrics. The codebase is available in [16]. The user provides a dataset and a configuration file, in which one specifies the desired features for new datasets. The script gives the user the desired datasets and, after training models, another script calculates evaluation metrics.

For sparse annotation datasets, the user can determine a slice step, a case step, and an axis in which annotations are kept. For example, a user can determine that every third slice for every fourth case in the axial direction is kept, and other labels are ignored in the training. The user can also add a secondary axis with a secondary slice step if one wants the dataset to be sparsed in multiple directions. For example, every eighth slice in the axial and coronal direction is kept, and others are ignored.

For datasets with added annotation noise, the user can determine what percentage of cases the noise is added to. The user can also determine parameters for elastic warp, magnitude of warp (α), and granularity (σ). Based on the given parameters, the code performs warping with Python’s elastic warp 3D method.

The script generates a folder that contains the baseline, the wanted sparse annotated, and the noise-added datasets. It also adds an ignore label to the sparse annotated datasets’ dataset.json file if it is not there already. The results script calculates Dice and HD95, as well as precision, recall, and volumetric ratio, for all datasets and prints them to a JSON file and a CSV table. It also makes bar charts of Dice and HD95 values across the datasets. Reporting precision, recall, and the predicted-to-true volumetric ratio provides immediate diagnostic evidence of systematic bias toward over- or under-segmentation, streamlining error analysis and guiding subsequent model calibration and refinement.

Sparse annotation

The impact of sparse annotation was researched with five different configurations, shown together in Table 2. Datasets were labeled with letters. In three of the sparse annotation datasets, 25% of the voxels are annotated, and in two, 50% of the voxels are annotated. All five datasets are sparse in the axial direction.

Table 2: Configurations of different sparse annotated datasets.

dataset id	slice step	case step	% of annotated voxels
A	4	1	25
B	1	4	25
C	2	2	25
D	2	1	50
E	1	2	50

In dataset A, every fourth slice is annotated. In dataset B, every fourth case is fully annotated, and in dataset C, every second slice is annotated for every second case. In dataset D, every second slice is annotated in all cases, and in dataset E, every second case is fully annotated. With these datasets, one can explore how the number of annotated voxels, as well as annotation strategies, impact models’ segmentation performance.

Annotation noise

The last dataset will serve as an indicator of the impact of annotation noise. Again, the same training set was used, and systematic noise was added to the annotations using the code described in 3.1. Noise was added to all cases, and parameters $\alpha = 50$ and $\sigma = 3$ were chosen based on [8], simulating a rushed annotator. An example of a made label is shown in Fig. 1.

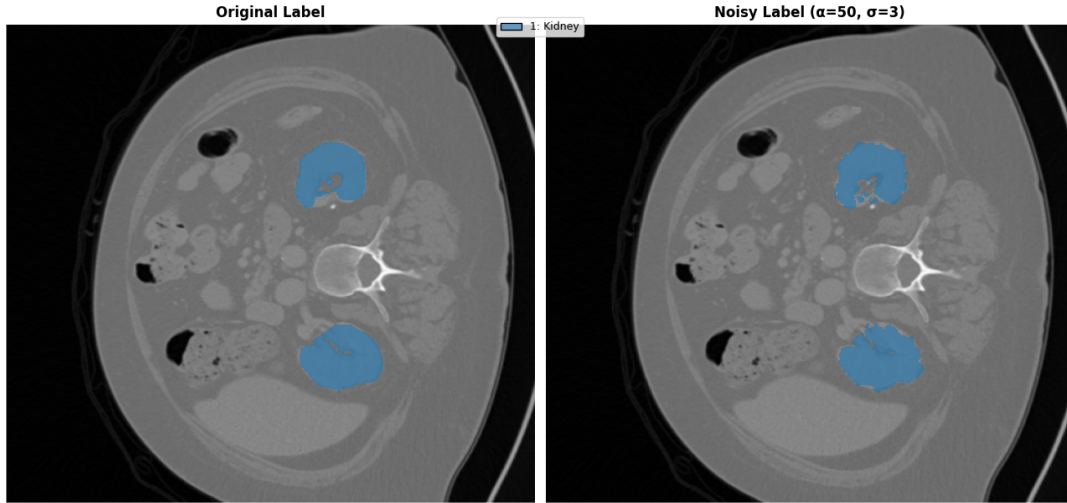


Figure 1: Example of noisy label made by warping shown beside the original label.

3.2 Methods

The nnU-Net was chosen as a training environment for its current state-of-the-art status in medical image segmentation. For computational resources, the Aalto University’s high-performance computing cluster, Triton, was used.

For computational efficiency, only a 3D full-resolution configuration was trained in each model. Validation was done by 5-fold cross-validation, and an ensemble model of each five was used for testing.

Evaluation metrics

The results program was used to get evaluation metrics. The Dice and HD95 means and medians were calculated for each dataset using Python’s medpy package. The program also calculates precision, recall, and volumetric ratio, but since no models were clearly over- or undersegmenting by these metrics, only Dice and HD95 were chosen to compare model performance. Means and medians were presented for each metric due to the different information that each metric tells. Worse means are usually due to several extreme cases, and medians indicate the model’s usual performance. In addition to calculated metrics, one randomly chosen segmentation of each model is shown for qualitative evaluation.

The use of AI

AI was used to generate and debug code for making the sparse annotation and annotation noise datasets. Also, an AI-powered writing assistant, Grammarly, was used to correct the language. All generated code and text were reviewed and edited by the author.

4 Results

The obtained Dice scores and HD95s are shown in Table 3. For easier comparison, values are also shown in bar charts. The dice score means, and medians in Fig. 2 and HD95 means in Fig. 3, and medians in Fig. 4. One segmentation slice for each model, as well as the original scan and ground truth, is shown in the Fig. 5.

Dataset Scenario					Dice		HD95 (mm)	
					Mean	(Median)	Mean	(Median)
Baseline					0.972	(0.980)	2.43	(0.98)
Sparse-Annotated	Dataset id	Slice step	Case step	% of annotated voxels				
	A	4	1	25 %	0.963	(0.975)	9.88	(0.98)
	B	1	4	25 %	0.934	(0.958)	71.99	(2.86)
	C	2	2	25 %	0.947	(0.968)	11.15	(1.37)
	D	2	1	50 %	0.961	(0.975)	11.91	(0.98)
	E	1	2	50 %	0.947	(0.974)	30.34	(1.36)
Annotation noise								
Noisy Dataset					0.961	(0.969)	16.74	(2.84)

Table 3: The mean and median Dice similarity coefficients and the 95th percentile Hausdorff distances for each scenario.

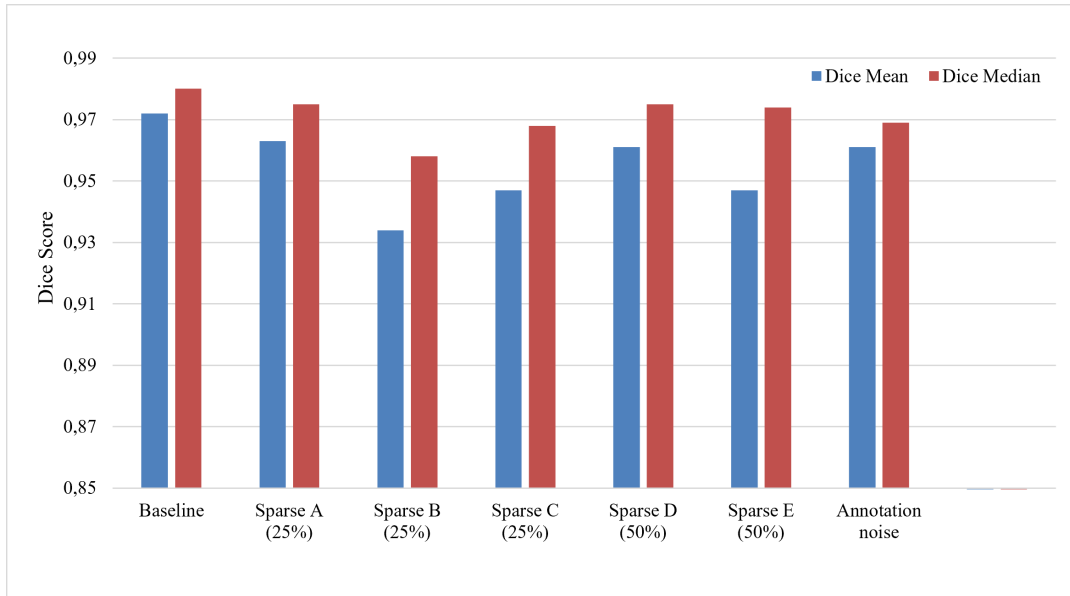


Figure 2: The dice coefficients means and medians for all scenarios.

In Table 3, one can see that Dice scores for all scenarios are over 0.93, so all models perform well. In Fig. 2, one can observe that values drop lowest with two

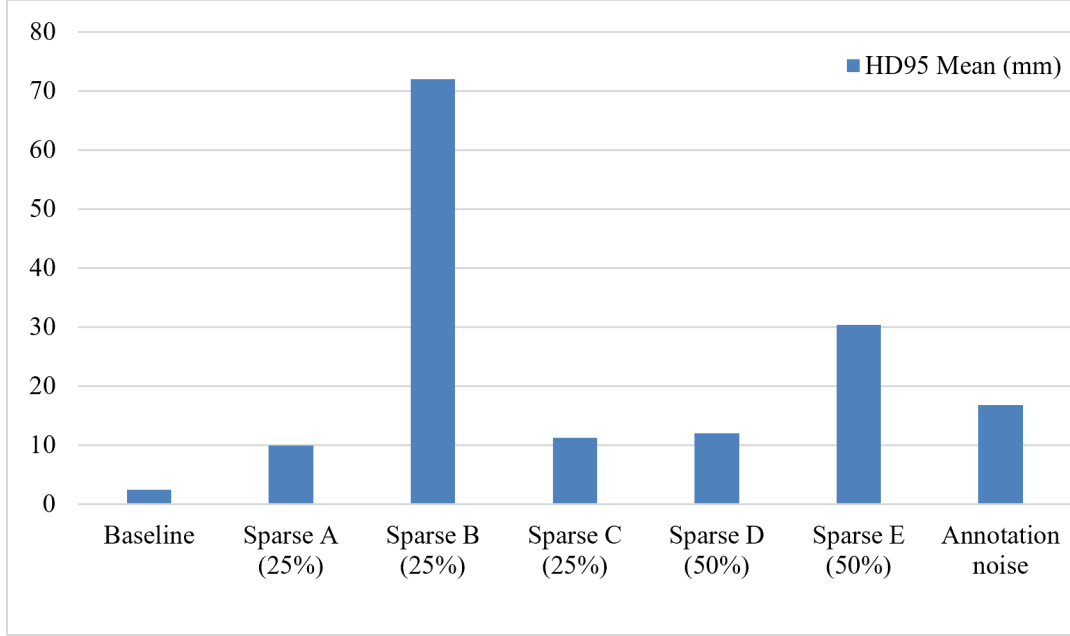


Figure 3: The HD95 means for all scenarios.

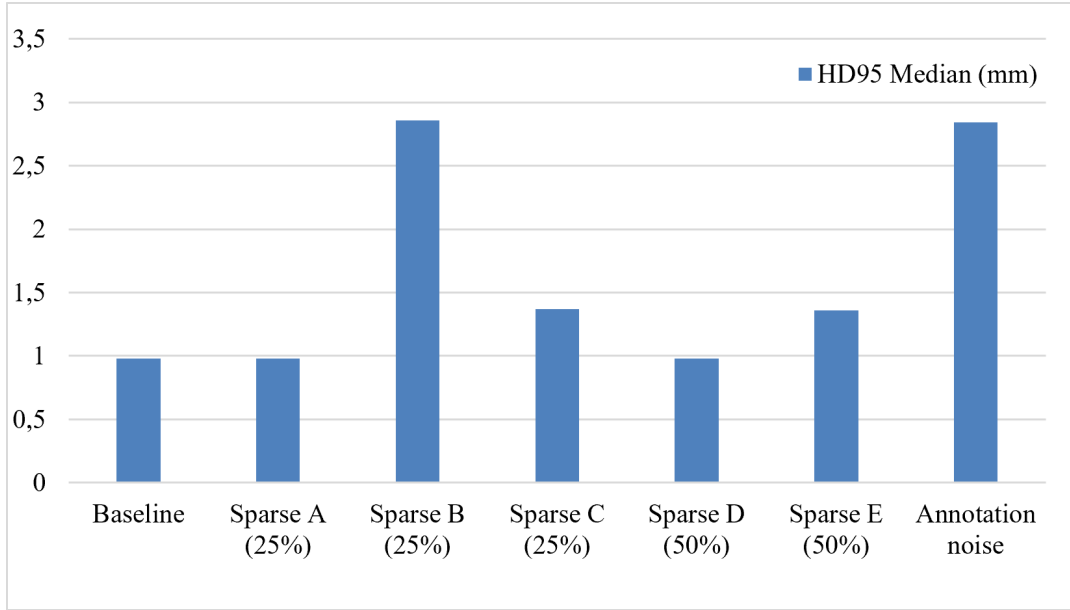


Figure 4: The HD95 medians for all scenarios.

scenarios where only 25% of voxels are annotated, but the difference between all sparse-annotated sets is less than 0.04. The annotation noise Dice score aligns with the best sparse-annotated Dice scores.

In bar charts 3 and 4, as well as by examining the segmentation boundaries in Fig. 5, one can clearly see the differences between the scenarios. In particular, the differences between the means of HD95 are large, most likely due to false positives. That can be clearly seen in Fig. 5 in sparse-annotated scenarios, where the model, in

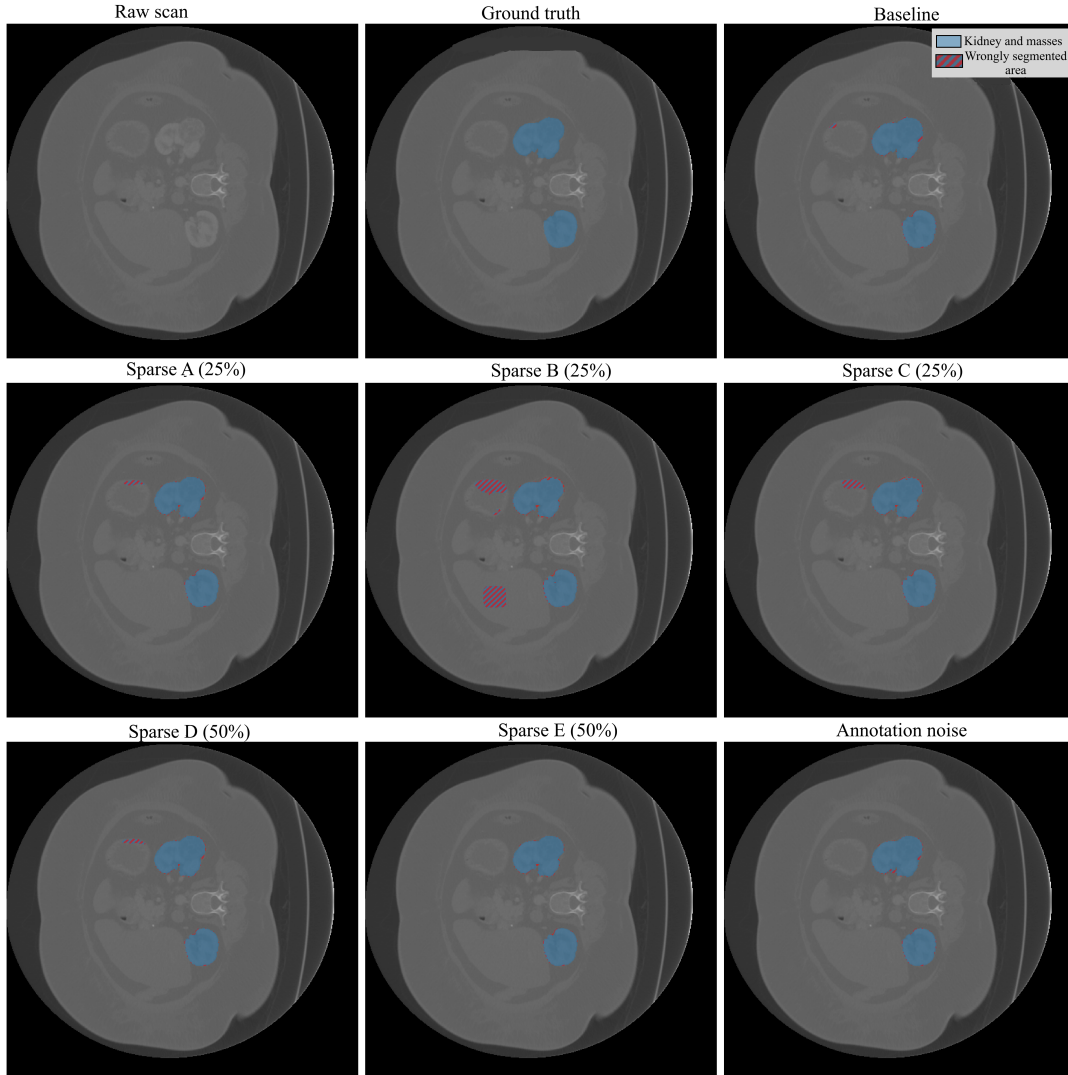


Figure 5: Raw scan, ground truth, and segmentations of each model for the same cases' same slice. Segmentations drawn on top of the scan using blue color, and wrongly segmented areas marked with red stripes.

addition to correct segmentations, draws segmentations on the left side of the kidney. Since the median HD95s in the Table 3 are quite small, the model does not segment those false positives in every case, but sometimes, as shown in the Fig. 5. Looking at the annotation noise impact on HD95 values in Fig. 3 and 4, there is a clear difference compared to the baseline, and both mean and median values are bigger than most of the sparsely annotated cases. In Fig. 5, there are no clear false positives to be seen in the noisy dataset.

5 Conclusions

The objectives of this study were to develop a Python-based framework for flexible configuration of datasets with controlled annotation sparsity and label noise, and to systematically evaluate their effects on medical image segmentation performance. The code environment is available [16], and in addition to dataset creation, there is also a results program that calculates and combines metrics and draws bar charts of Dice and HD95. For dataset creation, some options were not used in this research, but can be used in the future, such as choosing sparsifying to happen on different axes or a combination of axes, and modifying the amount of noise added to cases.

As detailed in Section 4, all trained models demonstrated strong performance according to the Dice coefficient. However, examination of the 95th-percentile Hausdorff distance (HD95) reveals that models trained on sparsely annotated datasets tend to achieve lower and thus superior HD95 values compared with models trained on datasets augmented with annotation noise, indicating tighter boundary agreement despite reduced annotation density. With sparse annotations, there are false positives to be seen, especially with sets where only 25% of voxels were annotated. But based on the mean HD95, models trained with label-noise-augmented data exhibit blurrier and less precise boundary delineations than those trained with sparsely annotated datasets.

Numerous avenues remain for further investigation in this area. In the present study, the evaluation was limited to segmenting a single large anatomical structure, and under this setting all models demonstrated strong performance. The impact should also be researched with harder, smaller, and more varied objects, such as tumors or cysts. In addition, the models should be tested on other datasets for other organs to verify the results. Moreover, there are multiple ways to annotate sparsely. In this study, sparsification was applied exclusively along the axial plane. It remains to be determined whether similar effects arise when sparsity is imposed in the sagittal or coronal planes, or through multi-plane combinations. Annotation noise was simulated in this study by a computer using elastic warping, but the results should be verified with a manually annotated set done by the rush annotator.

References

- [1] T. C. Hollon et al., “Near real-time intraoperative brain tumor diagnosis using stimulated raman histology and deep neural networks”, *Nature Medicine*, vol. 26, no. 1, pp. 52–58, 2020. DOI: [10.1038/s41591-019-0715-9](https://doi.org/10.1038/s41591-019-0715-9).
- [2] H. J. Aerts et al., “Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach”, *Nature Communications*, vol. 5, no. 1, p. 4006, 2014. DOI: [10.1038/ncomms5006](https://doi.org/10.1038/ncomms5006).
- [3] H. Cai, L. Qi, Q. Yu, Y. Shi, and Y. Gao, “3d medical image segmentation with sparse annotation via cross-teaching between 3d and 2d networks”, Tech. Rep., 2023. DOI: <https://doi.org/10.48550/arXiv.2307.16256>.
- [4] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, “3d u-net: Learning dense volumetric segmentation from sparse annotation”, Jun. 2016. [Online]. Available: <http://arxiv.org/abs/1606.06650>.
- [5] H. Su et al., “Sparsely annotated medical image segmentation via cross-sam of 3d and 2d networks”, Tech. Rep. [Online]. Available: <https://github.com/CTSegPilot/SA-Net.git>.
- [6] J. Stein, M. Di Folco, and J. A. Schnabel, “Sparse annotation strategies for segmentation of short axis cardiac mri”, in *Statistical Atlases and Computational Models of the Heart. Regular and CMRxRecon Challenge Papers*, O. Camara et al., Eds., Cham: Springer Nature Switzerland, 2024, pp. 66–76, ISBN: 978-3-031-52448-6. DOI: [10.1007/978-3-031-52448-6_7](https://doi.org/10.1007/978-3-031-52448-6_7).
- [7] S. Yu et al., “Robustness study of noisy annotation in deep learning based medical image segmentation”, *Physics in Medicine & Biology*, vol. 65, no. 17, p. 175 007, Aug. 2020, PMID: 32503027; PMCID: PMC7567130. DOI: [10.1088/1361-6560/ab99e5](https://doi.org/10.1088/1361-6560/ab99e5).
- [8] Y. Kim et al., “Unlocking robust semantic segmentation performance via label-only elastic deformations against implicit label noise”, *arXiv preprint arXiv:2508.10383*, 2025. DOI: [10.48550/arXiv.2508.10383](https://doi.org/10.48550/arXiv.2508.10383). arXiv: [2508.10383](https://arxiv.org/abs/2508.10383) [cs.CV].
- [9] F. Isensee, P. F. Jaeger, S. A. Kohl, J. Petersen, and K. H. Maier-Hein, “Nnu-net: A self-configuring method for deep learning-based biomedical image segmentation”, *Nature Methods*, vol. 18, no. 2, pp. 203–211, 2021. DOI: [10.1038/s41592-020-01008-z](https://doi.org/10.1038/s41592-020-01008-z).
- [10] K. Gotkowski, K. H. Maier-Hein, and F. Isensee, “Revisiting 3d medical scribble supervision: Benchmarking beyond cardiac segmentation”, *arXiv preprint arXiv:2403.12834*, 2024. DOI: [10.48550/arXiv.2403.12834](https://doi.org/10.48550/arXiv.2403.12834). arXiv: [2403.12834](https://arxiv.org/abs/2403.12834) [cs.CV].
- [11] M. Gunawardhana, F. Xu, and J. Zhao, “How good nnu-net for segmenting cardiac mri: A comprehensive evaluation”, *arXiv preprint*, 2024. DOI: [10.48550/arXiv.2408.06358](https://doi.org/10.48550/arXiv.2408.06358).

- [12] Y. M. Wong et al., “Machine learning prediction of dice similarity coefficient for validation of deformable image registration”, *Intelligence-Based Medicine*, vol. 10, p. 100 163, Jan. 2024, ISSN: 2666-5212. DOI: [10.1016/J.IBMED.2024.100163](https://doi.org/10.1016/J.IBMED.2024.100163).
- [13] T. Eelbode, J. Bertels, M. Berman, D. Vandermeulen, P. Maes, and D. Cremers, “Optimization for medical image segmentation: Theory and practice when evaluating with dice score or jaccard index”, *IEEE Transactions on Medical Imaging*, vol. 39, no. 11, pp. 3679–3690, Nov. 2020. DOI: [10.1109/TMI.2020.3002417](https://doi.org/10.1109/TMI.2020.3002417).
- [14] A. A. Taha and A. Hanbury, “Metrics for evaluating 3d medical image segmentation: Analysis, selection, and tool”, *BMC Medical Imaging*, vol. 15, p. 29, 2015. DOI: [10.1186/s12880-015-0068-x](https://doi.org/10.1186/s12880-015-0068-x). [Online]. Available: <https://doi.org/10.1186/s12880-015-0068-x>.
- [15] N. Heller, F. Isensee, A. Wood, et al., *Kits23: The 2023 kidney and kidney tumor segmentation challenge*, <https://kits-challenge.org/kits23/>, 2023.
- [16] E. Kuutti, *NnUnet-helper-sparseandnoise*, <https://github.com/EmmaKuutti/nnUnet-helper-sparseandnoise>, GitHub repository, 2026.